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Academic Year: 2022-23

Programme: B. Tech (CSBS) Year: II Semester: III

Subject: Object Oriented Programming

Date: 21 November 2022

Marks: 100
Time: 10.00 am - 1.00 pm
Durations: 3 (Hrs)
No. of Pages: 02

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO-1 BL-2	a.	Explain the concept of references in C++. Also elaborate the differences between pointers and references. Give an example.	[05]
CO- 1 BL-2	b.	Explain Data hiding with the help of an example.	[05]
CO-2 BL-1	c.	List Which binary operators can be overloaded? Write a small code snippet for any one operator explaining the same .	[05]
CO-4 BL-2	d.	List the basic building blocks of UML. Explain any one of them	[05]
Q2 CO-1 BL-6	a	Write a program to implement following: a. Create structure to manage the Employee data item and functions. I. Name of the Employee ii. Employee Number iii. Salary of Employee IV Designation of Employee Implement following functions to perform operations 1. Read the data of 3 Employees 2. Display the data of Employees 3. Search the specified Employee by name, if it is available or not, if available then display student record. 4 Search the specified Employee by employee number, if it is available or not, if available then display employee record	[10]

CO-3 BL-2,3	b	Explain the role of library functions. List four header files, their description and usage.	[10]
Q3 CO-2 BL-2,3	a	Elaborate compile time polymorphism. Demonstrate the same using code. Also mention advantages of polymorphism.	[10]
CO-3 BL-2, 6	b	What is the need of template function? Write a sort function to sort integer and floating type array using template function	[10]
Q4 CO-2 BL-2, 6	a	Define Friend class and friend function. Write a program showing the usage of friend class to elaborate the explanation. List 3 advantages of using friend keyword.	[10]
CO-4; BL-6	b	Design class diagram for Hospital Management system with detailed explanation of each component and cardinalities	[10]
Q5 CO-2 BL-3	a	How can memory be allocated and deallocated during program execution? Illustrate memory allocation for two integer values and compute the sum of these.	[10]
CO-4; BL-6	b	Design Activity diagram for ATM Management System with swimlanes	[10]
Q6 CO-2 BL-6	a	Create a class user _registration with data member as username and password accept the values in function acceptdata() while accepting the username check the constraints that it should accept 2 special characters in username and while accepting password check the constraints that it should be minimum of 8 in length . Create a function check() which should ask user to enter the username and password for login while logging in the username and password should be same as that of user_registration if it matches display "Logged in Successfully" else display "Login denied	[10]
CO-2 BL-1 2 6	b	What are constructors? Explain in detail. List advantages of using constructor over regular methods. Demonstrate using code	[10]
Q7 CO-2 BL-2	a	What do you infer from virtual function? Write syntax for the same. Also, discuss various rules pertaining to it.	[05]
CO-2 BL-2	b	Define encapsulation. Explains it with a real time example.	[05]
CO2 BL3	c	Write a program depicting "polymorphism through dynamic binding". Also illustrate its usage.	[05]
CO-2 BL-2	d	Explain scope resolution operator with the help of an example.	[05]